DDD-P-47C
January 13, 1976
SUPERSEDING
Int. Fed. Spec. DDD-P-0047b(DSA-DM)
December 11, 1972 and
Fed. Spec. DDD-P-47
April 29, 1959

FEDERAL SPECIFICATION

PAD, BED LINEN PROTECTIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers disposable, absorbent, hospital bed underpads, which afford protection to bed linens.

1.2 Classification. Bed linen protective pads shall be of the following types and sizes, as specified:

Type I 17 by 23-1/2 inches
Type II 22-1/2 by 23-1/2 inches

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6530

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standard:

T-403 - Bursting Strength of Paper.
T-404 - Tensile Breaking Strength of Paper and Paperboard.
T-410 - Basis Weight of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341.)

3. REQUIREMENTS.

3.1 Preproduction sample. Unless otherwise specified before production is commenced, a sample of the pads on contract and each component part thereof shall be made ready for the contracting officer or his authorized representative to examine and test to determine compliance with this specification (see 6.2). Approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility from compliance with all applicable provisions of this specification.

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3.2 Material. Material shall be of the highest quality throughout and suitable in all respects for the purpose intended.

3.2.1 Pad facing. The pad facing shall be made from long-fibered chemical wood pulp, specially treated to give a high wet strength. As an alternate, the facing shall be non-woven fabric of all cotton, all rayon, mixture of cotton and rayon, mixture of rayon and polyester, or mixture of cotton, rayon, and polyester, and suitable non-water-soluble binder. The facing shall be white or tinted bleedproof light blue. The facing shall be highly permeable, and when tested as specified in 4.4.1, a drop of distilled water shall be absorbed into the pad within 3 seconds. The facing shall meet the requirements specified in table I, when tested as specified in 4.4.2.

TABLE I. Pad facing.

Characteristic	Requirement
Basis weight (minimum)	
Wood pulp type	6.5 pounds per 480 24-by 36-inch sheets.
Cotton or rayon type	150 grains per square yard.
Tensile breaking strength, (dry), pounds per inch width, minimum	
Machine direction	3.0
Cross direction	0.25
Bursting strength, (wet), pounds per square inch, minimum	5.0

3.2.2 Absorbent material. The absorbent layers shall consist of one of the following:

a. Cellulose fluff between two layers of absorbent cellulose wadding.
b. Fluffed pulp absorbent material enveloped between layers of wet strength tissue.
c. Multiple layers of absorbent wadding.
d. A layer of absorbent fluff between a layer of absorbent wadding and polyethylene film backing with the fluff being secured to the film backing by means of a waffle weave pattern of adhesive on the film. The absorbent layers shall be overall embossed to provide uniform fluid dispersion.
e. A layer of absorbent fluff between a layer of absorbent wadding and polyethylene film backing with the fluff secured at the cut ends to the film backing by means of adhesive on the film. The absorbent layers shall be overall embossed to prevent the fluff from shifting and to provide uniform fluid dispersion.

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The abosrbent fluff shall be a uniform continuous web of macerated, absorbent, bleached, chemical pulp free of foreign matter. The abosrbent fluff shall be a uniformly distributed continuous layer with no thin or open spaces visible to the naked eye. The absorbent layers shall be completely submerged within 11 seconds and shall absorb not less than 12 times their own weight of water when tested as specified in 4.4.3.

3.2.3 Backing. Backing shall be color-pigmented polyethylene film not less than 0.75 mil thick.

3.3 Design and construction. The pad shall be an absorbent, soft, durable, disposable, hospital bed underpad suitable for protecting bed linens in medical treatment facilities. It shall consist of layers of absorbent material between a polyethylene backing and a pad facing. The construction of the pad shall be one of the following styles:

Style 1. Style 1 shall have absorbent material consisting entirely of multiple layers of cellulose wadding meeting the requirements specified in 3.2.2. The polyethylene backing shall be continuously sealed to the long sides of the facing of the pad. There shall be no breaks or openings in the seal. The ends of the cellulose wadding need not be sealed.

Style 2. Style 2 shall have absorbent material consisting of enveloped cellulose fluff or enveloped fluffed pulp meeting the requirements specified in 3.2.2. The polyethylene backing shall be sealed to all sides of the facing, providing a continuous seal around the periphery of the pad so there shall be no breaks or openings in the seal.

Style 3. Style 3 shall have absorbent material consisting of enveloped cellulose fluff or fluffed pulp meeting the requirements specified in 3.2.2. The polyethylene backing shall be continuously sealed to the long sides of the facing. There shall be no breaks or openings in the seal. The end of the absorbent layers shall be completely sealed to the ends of the polyethylene backing, so that there are no breaks or openings in the adhesion.

3.3.1 Dimensions. Sealed sides shall be soft and pliable, so as not to irritate sensitive areas of the patient's body. The pad shall have the following minimum dimensions:

TABLE II. Dimensions

	Dimensions (inches)	
	Type I	Type II
Absorbent layer	1(by 21	21 by 21
Finished pad	17 by 23-1/2	22-1/2 by 23-1/2

The absorbent materiel shall not extend beyond the ends of the backing.

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3.3.2 Count. For Type I, the average number of pads per package shall be not less than 200, with no less than 196 pads in any individual package. For Type II, the average number of pads shall be not less than 150, with no less than 147 pads in any individual package.

3.4 Flammability. When tested as specified in 4.4.4, the pad shall not support combustion and the burned area shall not reach the edge of the sample.

3.5 Workmanship. Workmanship shall be first class throughout. The finished pads shall be clean, well made, and free from any defects which may affect their appearance or serviceability. Pads shall be smooth and flat, and the absorbent layers shall be uniformly distributed throughout.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table III. The unit of product for inspection purposes shall be the individual pad (except where otherwise specified). Sample units shall be selected at random in such a manner that not more than 10 individual pads shall be selected from any one box. Pads shall be selected randomly from the package, and packages shall be selected randomly from the lot.

TABLE III. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	I	2.5
Minor defects	I	6.5
For dimensional examination	S-1	2.5

4.2.2 For tests.

4.2.2.1 Pad. Sampling for tests of the pad shall be in accordance with MIL-STD-105 and table IV. The acceptance number shall be zero for all sample sizes.

TABLE IV. Sampling for tests

Component	Characteristic	Requirement	Test procedure	Inspection level
Facing material	Permeability	3.2.1	4.4.1	S-1
	Weight and Strength	3.2.1	4.4.2	S-1
Absorbent layers	Absorbency	3.2.2	4.4.3	S-1
Pad	Flammability	3.4	4.4.4	S-1

4.2.2.2 Component materials. Sufficient quantity of the component materials shall be selected from each lot of material used in the manufacture of the pads to perform tests as required by the applicable subsidiary specifications or by tests specified herein. The sample size for each test shall be as indicated for inspection level S-1 in MIL-STD-105. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each. Not less than three samples shall be taken, regardless of lot size. Lot sizes shall be expressed as rolls.

4.3 Examination. The pads shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examinations shall be conducted in accordance with the classification of defects listed in table V. Examination shall not be restricted to the classified possible defects listed below.

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TABLE V. Classification of defects

Categories	Defects
Major	
101	Pad facing or water repellent backing of finished pad not free of tears, cuts, holes, or other damage.
102	Pads not complete with specified component parts.
103	Absorbent material not free of lumps or voids.
104	Adhesion of polyethylene backing to pad facing not continuous and free of breaks or openings.
105	Absorbent material not free of shifting within pad.
106	Exposed absorbent material not flush with or within open edges of pad.
107	Ends of absorbent material not completely anchored to polyethylene backing (applicable to Style 3).
108	Ends of absorbent material not free of tears or delamination (applicable to Styles 1 and 3).
109	Sealing edge not properly aligned.
Minor	
201	Component parts in pad not clean and free from stains, slime spots, or foreign matter.

4.3.2 Dimensional examination. The pads shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Permeability. A drop of distilled water shall be placed on the surface of the pad facing, backed by the remainder of the pad. A ring-type weight may be used to assure definite contact between the pad facing and the remainder of the pad.

4.4.2 Weight and strength. Tests on the facing material shall be conducted in accordance with the applicable method of the Technical Association of the Pulp and Paper Industry (TAPPI) as listed in table VI.

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TABLE VI. Weight and strength

Characteristic	Requirement paragraph	TAPPI Test method
Basis weight	3.2.1	T410
Tensile breaking strength, dry	3.2.1 1/	T404
Bursting strength, wet	3.2.1 2/	T403

1/ Distance between jaws at start of test shall be 3 inches.

2/ Four pieces of the pad facing, 6 inches by 6 inches, with the grain of the sheets matched in one direction, shall be attached by means of a spring paper clip on one edge, and then submerged in water at room temperature for 2 minutes. The pad facing shall be removed from the water and permitted to drain for 15 seconds, at the expiration of which time the four layers as a unit, shall be inserted in the bursting strength machine. Three readings shall be taken on each such specimen.

4.4.3 Absorbency. Ten specimens of the absorbent layers, each approximately 2-1/2 inches square, each weighing approximately 5 grams, shall be accurately weighed. Each specimen shall be laid on a flat 4 inch square of brass screen which has been previously weighed. The screen shall be held horizontally and about 1/2 inch above the surface of distilled water at 25° C. ± 1° C. The screen and specimen shall be lowered into the water, keeping the screen horizontal and low enough to prevent retardation of absorption but high enough to prevent disintegration of the cellulose as it expands due to absorption. The time elapsed after the specimen touches the water until it is completely submerged shall be taken, using a stop watch or pendulum. The screen and specimen shall be removed from the water and shall be allowed to drain for 10 seconds while in a horizontal position. The specimen and screen shall be lowered into a suitable tared receptacle and weighed to the nearest 0.1 gram. The average weight of water retained, based on the results of the 10 tests, shall be calculated.

4.4.4 Flammability. A specimen, 6 inches square, shall be taken from the finished pad for testing to determine compliance with the flammability requirement of 3.4. The specimen shall be dried for 1/2 hour at 140 degrees F. A burning paper match shall be dropped into the center of the specimen. The specimen shall not support combustion after the match goes out. The test shall be conducted in a draft free area.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table VII. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 expressed in terms of percent defective.

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TABLE VII. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.5.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be level A, B or C as specified (see 6.2).

5.1.1 Unit of issue. Unless otherwise specified, one package containing two hundred (200) pads of type I or one package containing one hundred and fifty (150) pads of type II, as specified, constitutes one unit of issue.

5.1.1.1 Unit package or shipping container.

5.1.1.1.1 Level A. Pad of one description only in quantities as specified in 3.3.1 shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-636, class weather resistant, grade V3c. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.1.1.1.2 Level B. Pad of one description only in quantities as specified in 3.3.1 shall be packaged as specified in 5.1.1.1.1, except the box shall conform to PPP-B-636, type CF, class domestic. Waterproofing and strapping shall not be required.

5.1.1.1.3 Level C. The subject commodity shall be packaged in substantial commercial containers (export when applicable) of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

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5.2 Marking.

5.2.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.2.2 Military activities. Each unit package shall be marked as specified for the exterior container in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Pads are intended for use as hospital bed underpads for the protection of bed linens in medical treatment facilities.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When specific test facilities are required (see 4.1).
(d) When preproduction sample is not required (see 3.1); all Department of Defense (DOD) activities see 6.3.

6.3 First Article instructions for DOD activities. The Armed Services Procurement Regulations (ASPR) Section No. 1, Part No. 9 and Section VII relate to First Article approval. The paragraphs within this specification shall be deleted by DOD activities and the requirements of the ASPR included in the contract or purchase order.

6.4 This specification covers the following items shown in the Federal Supply Catalog.

NSN	Item	Type
6530-00-065-9598	Pad, Bed Linen Protective 200s:	I 17 x 23-1/2 inches
6530-00-226-0246	Pad, Bed Linen Protective 150s:	II 22-1/2 x 23-1/2 inches

6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

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